

Apex Inventory

UI Requirement Specification

Application Overview

Route: All routes

Document Version: 1.0.0

Project: ReactExample (Apex Store Inventory Management)

1. Application Summary

Apex Inventory (Apex Store v1.0.0) is a React-based inventory management application for retail store operations. It provides dashboard analytics, inventory CRUD, stock intake, category management with store layout mapping, and supplier replenishment order tracking. Data is persisted via a mock Express backend (mock-backend/server.js) using db.json.

2. Technology Stack

- Frontend: React 19, TypeScript, Vite, React Router DOM 7
- Charts: Recharts
- Icons: Lucide React
- Backend: Express.js mock API on port 5001
- API proxy: /api routes proxied to backend via Vite config

3. Route Map

Route	Page Component
/	Dashboard
/inventory	InventoryList
/add-stock	AddStock
/categories	Categories
/orders	PlaceOrder
*	Redirect to /

4. Core Data Models

4.1 InventoryItem

id, name, category, quantity, state (liquid|solid), expiryDate, lastStockIn, salesTrend[6], seasonalSales{summer,monsoon,winter,spring}

4.2 Category

id, name, description, aisle, section, shelf

4.3 Order

id, itemName, category, quantity, state, orderDate, expectedArrival, status (pending|delivered), price

5. API Endpoints Summary

Method	Endpoint
GET	/api/items
POST	/api/items
PUT	/api/items/:id/quantity
DELETE	/api/items/:id
GET	/api/categories
POST	/api/categories
DELETE	/api/categories/:id
GET	/api/orders
POST	/api/orders
POST	/api/orders/:id/deliver
DELETE	/api/orders/:id

6. Cross-Page Business Rules

- Low stock: quantity ≤ 5 (used in Dashboard, Layout, Inventory filters)
- Out of stock: quantity $=== 0$
- In stock: quantity > 5
- Item name matching is case-insensitive for merge/delivery operations
- Categories must exist before adding stock or placing orders for new products

7. Requirement Documents Index

- 01-Layout-Navigation.pdf - App shell and sidebar
- 02-Dashboard-Overview.pdf - Analytics landing page

- [03-Inventory-Items.pdf](#) - Stock list and management
- [04-Add-Stock.pdf](#) - Stock intake form
- [05-Categories-Layout.pdf](#) - Category CRUD and store map
- [06-Replenish-Orders.pdf](#) - Order placement and tracking